英文版 Method

1. Experimental Procedure

1.1 Sample Quality Control

Please refer to Novogene's QC report for methods of sample quality control.

1.2 Library Construction, Quality Control and Sequencing

The genomic DNA sample was fragmented into short fragments. These DNA fragments were then end-polished, A-tailed, and ligated with full-length adapters for Illumina sequencing before further size selection. PCR amplification was then conducted unless specified as PCR-free. Purification was then conducted through the AMPure XP system (Beverly). The resulting library was assessed on the AgilentFragment Analyzer System (Agilent) and quantified to 1.5 nM through Qubit (ThermoFisher Scientific) and qPCR.

The qualified libraries were pooled and sequenced on Illumina platforms, according tothe effective library concentration and data amount required.

2. Bioinformatic analysis

The raw sequences were analysed for information at the end of sequencing. The data quality was evaluated to determine whether it met the standard. If the data meets the standard, the samples will be tested for variants, including SNP, InDel, CNV, SV, and annotated with the corresponding variant information. If the data do not meet the quality standard, additional testing or library reconstruction is required according to the actual situation.

2.1 Data quality control

2.1.1 Raw data

The original fluorescence image files are transformed to short reads (Raw data) by base calling and these short reads are recorded in FASTQ (Cock et al., 2010) format, which contains sequence information and corresponding sequencing quality information.

2.1.2 Evaluation of data (Data quality control)

Sequence artifacts, including reads containing adapter contamination, low quality nucleotides and unrecognizable nucleotide (N), undoubtedly set the barrier for the subsequent reliable bioinformatics analysis. Hence quality control is an essential step and applied to guarantee the meaningful downstream analysis.
The steps of data processing were as follows:

(1) Discard a paired reads if either one read contains adapter contamination (>10 nucleotides aligned to the adapter, allowing ≤ 10% mismatches);
(2) Discard a paired reads if more than 10% of bases are uncertain in either one read;
(3) Discard a paired reads if the proportion of low quality (Phred quality <5) bases is over 50% in either one read.3 References

2.2 Reads Mapping to Reference Sequence

Valid sequencing data was mapped to the reference genome by Burrows Wheeler Aligner (BWA) (Li et al., 2009a) software to get the original mapping results stored in BAM format (parameter: mem -t 4 -k 32 -M). Then, the results were dislodged duplication by SAMtools (Li et al., 2009b) (parameter: rmdup) and Picard (http://broadinstitute.github.io/picard/).

2.3 Variant detection and annotation

2.3.1 SNP/InDel

The raw SNP/InDel sets are called by GATK(DePristo et al., 2011) with the parameters as ' – gcpHMM 10 -stand_emit_conf 10 -stand_call_conf 30'. Then we filtered this sets using the following criteria:

SNP: QD < 2.0, FS > 60.0, MQ < 30.0, HaplotypeScore > 13.0,

MappingQualityRankSum < -12.5, ReadPosRankSum < -8.0

INDEL: QD < 2.0, FS > 200.0, ReadPosRankSum < -20.0

2.3.2 CNV and SV

CNVs were detected by CNVnator (Abyzov et al., 2011) to acquire potential deletion and duplication (parameter: - call 100).

SVs were detected by BreakDancer(Chen et al., 2009).

2.4 Annotation

ANNOVAR (Wang et al., 2010) was used for functional annotation of variants. The UCSC known genes were used for gene and region annotations.

2.5 Gene function

We used three databases to predict gene functions. They were respective GO(Gene Ontology，http://geneontology.org/)、KEGG (Kyoto Encyclopedia of Genesand Genomes，http://www.genome.jp/kegg)、COG(Cluster of Orthologous Groupsof proteins，http://www.ncbi.nlm.nih.gov/COG/)

The basic steps for feature annotation are as follows:

1) BLAST alignment of predicted genes with functional databases (blastp, evalue $\leq$ 1e-5);

2) BLAST result filtering: For the BLAST results of each sequence, the comparison result with the highest score (default identity>=40%, coverage>=40%) is selected for annotation.

3 References

Abyzov A, Urban AE, Snyder M, Gerstein M. CNVnator: an approach to discover, genotype, and characterize typical and atypical CNVs from family and population genome sequencing. Genome Res. 2011;21(6):974-984. doi:10.1101/gr.114876.110

Chen K, Wallis JW, McLellan MD, et al. BreakDancer: an algorithm for high-resolution mapping of genomic structural variation. Nat Methods. 2009;6(9):677-681. doi:10.1038/nmeth.13635

Cock PJ, Fields CJ, Goto N, Heuer ML, Rice PM. The Sanger FASTQ file format for sequences with quality scores, and the Solexa/Illumina FASTQ variants. Nucleic Acids Res. 2010;38(6):1767-1771. doi:10.1093/nar/gkp1137

Li H, Durbin R. Fast and accurate short read alignment with Burrows-Wheeler transform.Bioinformatics.2009a;25(14):1754-1760.doi:10.1093/bioinformatics/btp324

Li H, Handsaker B, Wysoker A, et al. The Sequence Alignment/Map format and SAMtools.Bioinformatics.2009b;25(16):2078-2079.doi:10.1093/bioinformatics/btp352

DePristo M A, Banks E, Poplin R, et al. A framework for variation discovery and genotyping using next-generation DNA sequencing data. Nature Genetics. 2011, 43(5):491-498 .doi: 10.1038/ng.806

Wang K, Li M, Hakonarson H. ANNOVAR: functional annotation of genetic variants from high-throughput sequencing data. Nucleic Acids Res. 2010;38(16):e164. doi:10.1093/nar/gkq603